MIL-S-0010133D(EA)
29 October 1974
SUPERSEDING
MIL-S-0010133C(MU)
3 July 1968
USED IN LIEU OF
MIL-S-10133B
2 October 1956

MILITARY SPECIFICATION

SEAT, OUTLET-VALVE, ALUMINUM-BASE-ALLOY DIE-CASTING FOR OUTLET VALVE, C15

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-S-10133B. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of chemical-biological mask outlet valve seat.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-591 - Aluminum Alloy Die Castings.

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).
FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

FSC 4240

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

D5-2-606	-	Seat, Outlet Valve.
B5-2-1006	-	Disk, Outlet Valve.
DLB136-45-202-14	-	Indicator, Outlet Valve Leakage, M4A1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the seat shall conform to the specifications and drawings forming a part of this specification.

3.2 Manufacture. The seat shall be manufactured of aluminum alloy conforming to QQ-A-591 and as shown on Drawing D5-2-606.

3.3 Physical requirements.

3.3.1 Rigidity. The hub shall depress not more than 0.016 inch when tested as specified in 4.2.4.1.

3.3.2 Impact Resistance. The outlet valve seat shall not break, nor shall the ribs or the hub be depressed so that the distance from any portion of the upper surface of the hub to the upper edge of the sealing rim surface has been increased by more than 0.012 inch when tested as specified in 4.2.4.2.

3.3.3 Leakage. The leakage through the seat assembly, before and after corrosion testing, shall be no more than 15 milliliters of air per minute when tested as specified in 4.2.4.3.

3.3.4 Corrosion resistance. The seat shall show no corrosion when tested as specified in 4.2.4.4.

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3.4 Workmanship. The portions of the seat designed to come in contact with rubber parts shall be smooth and free of sharp edges. The seat shall be substantially free from flash.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A production lot shall consist of the outlet valve seats produced by one manufacturer from the same materials, at the same plant, and under the same manufacturing conditions under continuous production. Any shut down or any alteration in the continiuty of production, such as a change in machine, die, modification of die, surface treatment, or change or pot or composition of alloy, shall be considered as the start of a new lot. Die castings shall be kept separated by lot for trimming, finishing, and shipping. The inspection lot for nondestructive inspection, within the above limitations, shall be designated by the Government inspector.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with the classification of defects and MIL-STD-105.

4.2.2.2 For destructive tests.

(a) Rigidity and impact resistance. Sixty-six seats, 22 for test and 44 for retest, shall be taken at random from each lot for tests as specified in 4.2.4.1 and 4.2.4.2.

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(b) Corrosion resistance. Twenty-five seats which have passed the leakage test shall be taken for the corrosion resistance test as specified in 4.2.4.4.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive tests. Sample seats shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For destructive tests. Sample seats shall be tested in accordance with 4.2.4.1, 4.2.4.2, and 4.2.4.4.

4.2.3.3 Classification of defects.

(a) Seat, outlet valve (Dwg. D5-2-606).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.25 percent defective	
101	Leakage	4.2.4.3
Major:	AQL 0.65 percent defective	
102	Non-fills	
103	Axial thickness of hub at hole (0.150)	
104	Height of hub in relation to sealing rim surface (0.160)	
105	Hub hole diameter (0.176)	
106	Sealing rim surface rough, irregular, burred, or cracked	
107	Neck flange sharp edge	
108	Neck cracked	
109	Contamination (grease and oil)	
110	Hub cracked, burrs, breaks, rough or sharp edges	
111	Chamfer excessive at lower hub surface	
112	Diameter (1.130)	

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4.2.3.3 Classification of defects (continued).

(a) Seat, outlet valve (Dwg. D5-2-606).

Categories	Defects
Minor:	AQL 1.0 percent defective
201	Overall height
202	Cracks, burrs, breaks, rough or sharp edges (a) Cover spacing rim (b) Cover flange (c) Cover spacing boss (d) Cover securing boss (e) Neck flange (other than 107) (f) Ribs
203	Excessive mold flash
204	Neck outside diameter
205	Coating missing or inadequate
	AQL 2.5 percent defective
206	Contamination (other than 109)
207	Color
208	Identification marking missing or illegible

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Rigidity. Slowly apply a load of 50 pounds to the flat surface of the hub. Remove the weight after 1 minute and measure any depression with a micrometer.

4.2.4.2 Impact resistance. Drop a 1/4 pound cylindrical steel pellet, 1 inch in diameter with a spherical end, from a height of 24 inches onto the top of the hub. Observe the seat for compliance with 3.3.2.

4.2.4.3 Leakage. Assemble a dry disk (Dwg B5-2-1006) to a dry seat and test for leakage with the M4A1 Outlet Valve Leakage Indicator (DLB136-45-202-14).

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4.2.4.4 Corrosion resistance. Determine corrosion resistance in accordance with method 6061 of FED-STD-141. The test shall be run on the seats (after leakage tests) and the duration of the test shall be 120 consecutive hours. After examining for corrosion the seats shall be washed with tap water, dried, and retested for leakage (4.2.4.3).

4.2.5 Acceptance/rejection criteria.

4.2.5.1 Rigidity and impact resistance. If one or two seats fail the tests specified in 4.2.4.1 and 4.2.4.2 repeat the tests using the specimens reserved for retest (4.2.2.2(a)). Reject the lot if more than two failures are observed in test plus retest.

4.2.5.2 Corrosion resistance. If any specimen fails the corrosion test reject the lot represented.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking, interplant shipment (see 6.4). The seats shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with FED-STD-123 and shall include the contractors name, address, and contract number.

5.2 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by its National Stock Number.

6. NOTES

6.1 Intended use. This specification covers an outlet valve seat intended to be used as a component of the C15 Outlet Valve.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Outlet valve disks. The outlet valve disks for use in leakage testing will be furnished by the Government.

6.4 Interplant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor to a military arsenal or plant.

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6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 216. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.6 Figure 1. Figure 1 is a miniature reproduction of assembly drawing and is attached for information only.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A644

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THIS FIGURE IS A MINIATURE COPY OF AN EDGEWOOD ARSENAL DRAWING AND IS FOR INFORMATION ONLY. THE DRAWING REFERENCE IN PARAGRAPH 2.1 GOVERNS

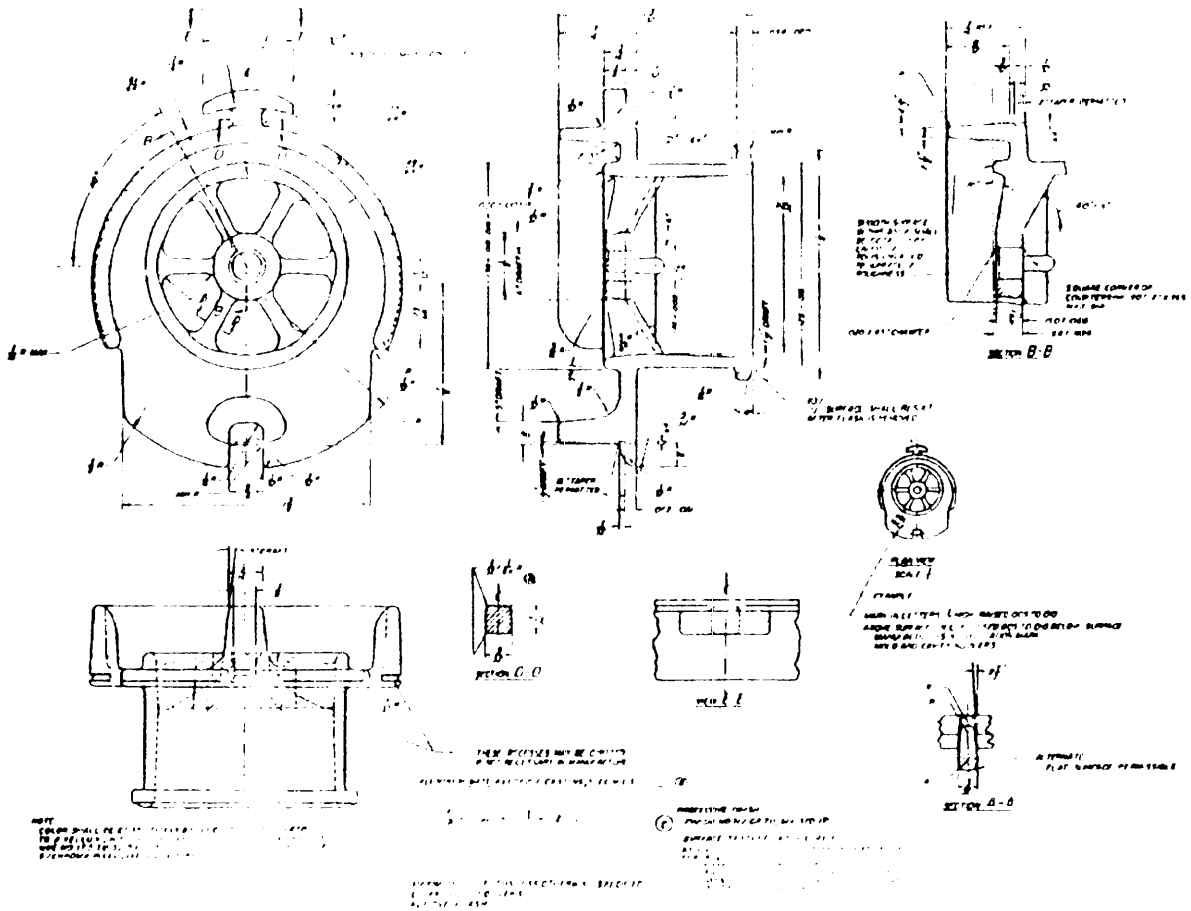


Figure 1

Seat, Outlet-Valve, Aluminum-Base-Alloy
Die-Casting for Outlet Valve, C-15

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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